Carleman Moments Do Not Recover an Unaugmented Path Law: A Disproof of Conjecture 00000004003 on Its Literal Process-Law Reading

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Abstract

This counterexample concerns the original, unaugmented process law, as stated in the bilingual source. It does not concern a law on signatures or on tree-like equivalence classes. Two smooth based one-dimensional closed paths have the same signature at every level, but different laws as processes on $[0,1]$. Both laws have finite support and satisfy an explicit divergent inverse-even-moment-root Carleman condition. Lean verifies genuine recursive iterated integrals, all signature levels, the moment condition, and inequality of the actual path laws on the observation interval.

Reading and essential boundary. Both source languages say that the expected signature determines the process distribution if and only if a Carleman condition holds. They do not specify a tree-like quotient, time augmentation, or a signature-injective class of paths. We refute the sufficiency direction for the literal original path distribution. This boundary matters: Hambly–Lyons identify signatures with tree-like equivalence classes of finite-length paths [1]; Chevyrev–Lyons study probability measures on signatures [2]. Our two paths are tree-like and have the same signature law. Thus this example does not contradict either of those determinacy frameworks. It does not address a corrected source interpretation in which "distribution" means a signature law or a law on tree-like equivalence classes, nor a time-augmented signature. A Carleman hypothesis cannot restore information that the unaugmented signature already discards deterministically.

The process family and its ordinary moment problem. For $a \in [1,2]$, set

$$x_a(t) = 4at(1-t), \qquad 0 \le t \le 1.$$

These are smooth paths with common starting point 0 and common endpoint 0. Their midpoint values are $x_a(1/2) = a$, and

$$\|x_a\|_\infty = a,$$

because $4t(1-t) = 1-(2t-1)^2 \in [0,1]$, with equality at $t = 1/2$. Consequently the family is injectively parametrized by amplitude, recoverable from the original process. The ordinary amplitude moment problem is a compactly supported scalar moment problem. We take

$$\mu = \delta_{x_1}, \qquad \nu = \delta_{x_2}$$

as Borel probability measures on $C([0,1],\mathbb{R})$. They are different: midpoint evaluation has law δ_1 under μ and δ_2 under ν.

Carleman convention and verification. The source does not specify the moment object in its Carleman condition. We use the explicit radial path-norm version

$$m_k(\eta) = \int \|p\|_\infty^k \, d\eta(p) < \infty, \qquad \sum_{n=1}^{\infty} m_{2n}(\eta)^{-1/(2n)} = \infty. \tag{1}$$

For the family above this is exactly the ordinary positive-amplitude Carleman condition. In particular,

$$m_{2n}(\mu) = 1, \qquad m_{2n}(\nu) = 2^{2n},$$

so the respective summands in (1) are 1 and 1/2. All moments are finite and all even moments are strictly positive.

This choice does not exploit heavy tails or an indeterminate moment problem. Any amplitude law supported in $[1, 2]$ has inverse-even-root terms at least 1/2. For each of our two Dirac laws, every fixed coordinate or finite collection of sampled coordinates is deterministic and bounded, hence satisfies the usual scalar or coordinatewise Carleman test, with the usual infinite inverse-root convention for a zero moment. The total variation of x_a is $2a$: the path rises from 0 to a and falls back to 0. Thus a variation-norm version also has constant positive inverse-root terms. We do not assert that there is one canonical infinite-dimensional Carleman definition; these observations make the chosen convention and the other common moment readings explicit.

Actual signatures at every level. For a smooth scalar path with $p(0) = 0$, define its iterated integrals recursively by

$$S_0(p; t) = 1, \qquad S_{n+1}(p; t) = \int_0^t S_n(p; s)p'(s)\, ds.$$

This is the one-dimensional tensor signature, with its unique scalar coordinate at each tensor level. For our paths, induction gives

$$S_n(x_a; t) = \frac{x_a(t)^n}{n!} \qquad (n \geq 0). \tag{2}$$

The base case is $S_0 = 1$. If the formula holds at n, then

$$\frac{d}{dt}\frac{x_a(t)^{n+1}}{(n+1)!} = \frac{x_a(t)^n}{n!}x_a'(t).$$

The fundamental theorem of calculus proves the next case, since $x_a(0)^{n+1} = 0$. All functions involved are polynomials and are integrable on every compact interval. At $t = 1$ the entire signature is therefore

$$S(x_a) = (1, 0, 0, \ldots),$$

independently of a.

Theorem 1. *There are two distinct smooth based process laws on* $[0, 1]$ *satisfying* (1) *whose expected signatures agree at every level. Thus the source's claimed sufficiency fails for original unaugmented process laws.*

Proof. Use μ and ν above. The Carleman verification and inequality of their midpoint laws were proved above. Integrating (2) against either Dirac law gives the same expected signature $(1, 0, 0, \ldots)$. □

Embedding the paths as $(x_a, 0, \ldots, 0)$ yields the same example in every positive dimension. Any tensor word containing another coordinate has zero iterated integral; the pure first-coordinate words were computed above. Hence the obstruction is not confined to dimension one. No separate assertion about an optimal truncation level on a signature-injective class is made here.

Formal verification and scope. Lean defines S_n by actual recursive Mathlib interval integrals involving the derivative of a continuous real-valued path. It proves (2) for every amplitude, level, and real upper endpoint using derivative rules and the fundamental theorem of calculus. It verifies smoothness, common starting point, midpoint recovery, the exact supremum norm, two genuine Dirac probability measures, integrability of all norm moments and signature coordinates, strict positivity of even norm moments, and nonsummability of the actual inverse-even-moment-root sequence. All expected-signature coordinates are equal. The final theorem negates a universally quantified Carleman-implies-determination proposition, including all these admissibility assumptions.

For derivative bookkeeping, paths are stored as global smooth extensions in $C(\mathbb{R}, \mathbb{R})$. Crucially, the conclusion compares their pushforward Borel laws on $C([0,1], \mathbb{R})$, and formal inequality is detected at the midpoint inside that interval. No disagreement outside the observation interval is used. The general-amplitude-law remark, variation-norm observation, higher-dimensional embedding, and tree-like interpretation are proved or explained here on paper rather than formalized. The formal result is the explicit scalar counterexample with the stated radial Carleman condition, not a refutation of a signature-law uniqueness theorem.

Source and reproduction. The exact bilingual source is `SOURCE.md`; its frozen commit and SHA-256 are recorded in `verification/SOURCE_PROVENANCE.md`. Use Lean 4.19.0 and the pinned Mathlib manifest. From `lean/`, run

```
lake env lean -DwarningAsError=true Main.lean
```

The strict log records the real compiler result. The final axiom audit uses only `propext`, `Classical.choice`, and `Quot.sound`. Development was by a delegated AI agent with self-review; the coordinating agent reviews it separately. No external independent review is claimed.

References

[1] B. Hambly and T. Lyons, *Uniqueness for the signature of a path of bounded variation and the reduced path group*, arXiv:math/0507536.

[2] I. Chevyrev and T. Lyons, *Characteristic functions of measures on geometric rough paths*, arXiv:1307.3580.